

Practical 3

Exp.name : Write a program in C++ to sort the numbers in an array using separate functions for read, display, sort and swap. The objective of this assignment is to learn the concepts of input, output, functions, call by reference in C++.

Name : Amar Sadashiv Jadhav

PRN : B25ET1090

Batch: SY2

Program:

```
#include <iostream>
using namespace std;

void inputArray(int a[], int n) {
    cout << "Enter " << n << " elements:\n";
    for (int i = 0; i < n; i++) {
        cin >> a[i];
    }
}

void displayArray(int a[], int n) {
    cout << "Array elements are: ";
    for (int i = 0; i < n; i++) {
        cout << a[i] << " ";
    }
    cout << endl;
}

void swapElements(int &x, int &y) {
    int temp = x;
    x = y;
    y = temp;
}

void sortArray(int a[], int n) {
```

```

    for (int i = 0; i < n - 1; i++) {
        for (int j = 0; j < n - i - 1; j++) {
            if (a[j] > a[j + 1]) {
                swapElements(a[j], a[j + 1]);
            }
        }
    }
}

```

```

int main() {
    int n;

    cout << "Enter the size of array: ";
    cin >> n;
    int a[n];
    inputArray(a, n);

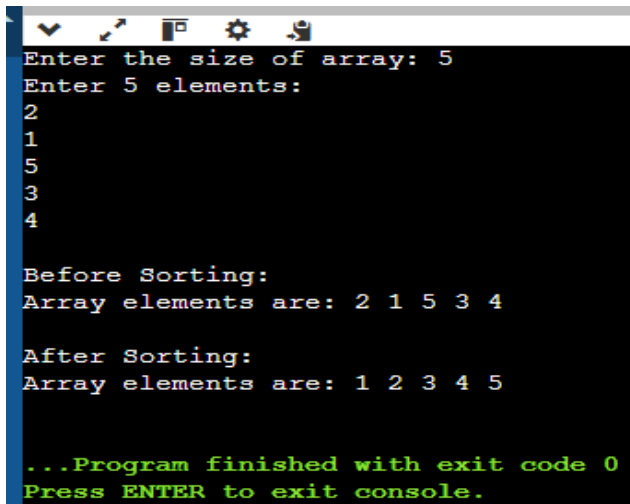
    cout << "\nBefore Sorting:\n";
    displayArray(a, n);
    sortArray(a, n);

    cout << "\nAfter Sorting:\n";
    displayArray(a, n);

    return 0;
}

```

Output :



```

Enter the size of array: 5
Enter 5 elements:
2
1
5
3
4

Before Sorting:
Array elements are: 2 1 5 3 4

After Sorting:
Array elements are: 1 2 3 4 5

...Program finished with exit code 0
Press ENTER to exit console.

```